– Not 3.5

Ch 3.3: Even More Linear Regression

Lecture 6 - CMSE 381

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Announcements

3.1, 3.2, 3.3

Last time:

- 3.2 Multiple Linear Regression

Announcements:

- Second homework due Friday!
  Covers:
  - Fri 1/21 Lecture
  - Mon 1/24 Lecture
  - Weds 1/26 lecture
- Office hours
- HW1 Grades in D2L

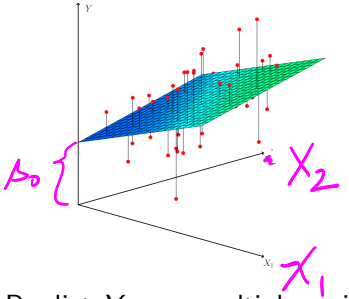
Covered in this lecture

- Qualitative predictors
- Extending the linear model with interaction terms
- Hierarchy principle
- Polynomial regression

Section 1

Review from last time

Linear Regression with Multiple Variables



- Predict Y on a multiple variables X

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p X_p + \varepsilon$$

- Find good guesses for $\hat{\beta}_0, \hat{\beta}_1, \cdots, \hat{\beta}_p$
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$

$= \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots \hat{\beta}_p x_p$

- $e_i = y_i - \hat{y}_i$ is the ith residual
- RSS $= \sum_i e_i^2$
- RSS is minimized at *least squares coefficient estimates*

$\underbrace{\hat{\beta}_0, \ldots, \hat{\beta}_p}_{\text{closed form}}$

Questions to ask of your model

1. Is at least one of the predictors $X_1, \cdots, X_p$ useful in predicting the response?
2. Do all the predictors help to explain Y, or is only a subset of the predictors useful?
3. How well does the model fit the data?
4. Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

Hypothesis test

F-stat.

$H_0: \beta_1 = \beta_2 = \cdots = \beta_p = 0$

H_a: at least one $\beta_j \neq 0$

Forward / Backward Selection

RSE

R^2

"Eyeball test"

Confidence intervals

prediction intervals

Spoiler: Turn them into quantitative

Section 2

Qualitative Predictors

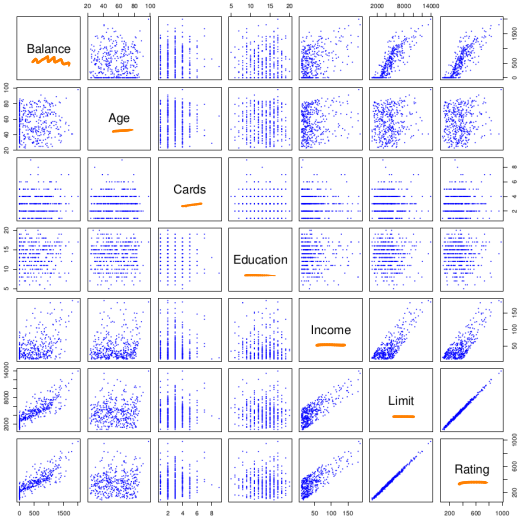
Classification vs Regression

↳ Y is Qualitative vs Quantitative

Quantitative predictors

Numerical valued inputs X_i

New data set! Credit card balance



Qualitative variables

- `own`: house ownership Yes/No
- `student`: student status Yes/No (?)
- `status`: marital status
- `region`: East, West, or South

What if....

... your variables aren't quantitative?

- Home ownership
- Student status
- Major
- Gender
- Ethnicity
- Country of origin

Qualitative predictors - One hot encoding

Y X

Ex. Investigate differences in credit card balance between people who own a house and those who don't, ignoring the other variables

Create a new variable

$$x_i = \begin{cases} 1 & \text{if } i\text{th person owns a house} \\ 0 & \text{if } i\text{th person does not own a house} \end{cases}$$

Model:

1 noise

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person owns house} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person doesn't own} \end{cases}$$

0

avg balance is $\beta_0 + \beta_1$

avg cc balance is β_0 don't own

β_1 = avg diff in balance

Interpretation

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	509.80	33.13	15.389	< 0.0001
`own[Yes]`	19.73	46.05	0.429	0.6690

Model:

$$y = \underbrace{509.80}_{\hat{\beta}_0} + \underbrace{19.73}_{\hat{\beta}_1} \cdot x_{owner}$$

Avg cc balance for non-owners is \$509.80

— " — owners is $\beta_0 + \beta_1 = 509.8 + 19.73$

$= \$529.53$

Who cares about 0/1?

$$x_i = \begin{cases} 1 & \text{if } i\text{th person owns a house} \\ -1 & \text{if } i\text{th person does not own a house} \end{cases}$$

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person owns} \\ \beta_0 - \beta_1 + \varepsilon_i & \text{if } i\text{th person doesn't own} \end{cases}$$

$$y = \underbrace{\$519.665}_{\hat{\beta}_0} + \underbrace{\$49.865}_{\hat{\beta}_1} \cdot x_i + \varepsilon_i$$

Same prediction y for inputs

Qualitiative Predictor with More than ~~Two~~ One Levels

≥ 2

Region: South, West, East

	x_{i1}	x_{i2}
South	1	0
West	0	1
East	0	0

Create spare dummy variables:

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person from South} \\ 0 & \text{if } i\text{th person not from South} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person from West} \\ 0 & \text{if } i\text{th person not from West} \end{cases}$$

East = baseline

- Always one fewer variable than levels
- Level w/ no dummy variable → baseline

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$$

$$\begin{cases} \beta_0 + \beta_1 \cancel{x_{i1}} + \varepsilon_i & \text{if } i\text{th person from South} \\ \beta_0 + \beta_2 \cancel{x_{i2}} + \varepsilon_i & \text{if } i\text{th person from West} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person from East} \end{cases}$$

More on multiple levels

$$\begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person from South} \\ \beta_0 + \beta_2 + \varepsilon_i & \text{if } i\text{th person from West} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person from East} \end{cases}$$

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	531.00 ($\hat{\beta}_0$)	46.32	11.464	< 0.0001
`region[South]`	−18.69 ($\hat{\beta}_1$)	65.02	−0.287	0.7740
`region[West]`	−12.50 ($\hat{\beta}_2$)	56.68	−0.221	0.8260

WARNING

- Avg balance for baseline (East) $= \hat{\beta}_0 = \$531$
- South has \$18.69 less than east (on avg)

  $\beta_0 + \beta_1 \rightsquigarrow$ South $\beta_0 + \beta_1 = \$531 - \$18.69$
  $= \$512.31$
- West has $\beta_0 + \beta_2 = 531 - 12.50 = \$518.50$

p-values for dummy variables large

can't reject null hypothesis

No statistical evidence that there's diff in balance by region

Section 3

Extending the linear model

Assumptions so far

Advertising

$$\hat{Y}_{sales} = \beta_0 + \beta_1 \cdot X_{TV} + \beta_2 \cdot X_{radio} + \beta_3 \cdot X_{newspaper}$$

Assumed (implicitly) that the effect on sales by increasing one medium is independent of the others.

What if spending money on radio advertising increases the effectiveness of TV advertising? How do we model it?

Interaction term

interaction term

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

$$\beta_1 \boxed{X_{TV}} + \beta_3 X_R \boxed{X_{TV}}$$

$$\begin{aligned} Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\ &= \beta_0 + (\beta_1 + \beta_3 X_{radio}) X_{TV} + \beta_2 X_{radio} + \varepsilon \end{aligned}$$

R^2 for this: 96.8% vs 89.7% w/ interaction term

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	6.7502	0.248	27.23	< 0.0001
`TV`	0.0191	0.002	12.70	< 0.0001
`radio`	0.0289	0.009	3.24	0.0014
`TV×radio`	0.0011	0.000	20.73	< 0.0001

Strong evidence for $H_a: \beta_3 \neq 0 \Leftarrow$ low interaction term p-value

Interpretation

$$
\begin{aligned}
Y_{sales} &= \beta_0 + \beta_1 X_{TV} + \beta_2 X_{radio} + \beta_3 X_{radio} X_{TV} + \varepsilon \\
&= \beta_0 + \underbrace{(\beta_1 + \beta_3 X_{radio})}\underbrace{X_{TV}} + \beta_2 X_{radio} + \varepsilon
\end{aligned}
$$

X_{TV} ↑ \$1000

	Coefficient	Std. error	t-statistic	p-value
Intercept	6.7502	0.248	27.23	< 0.0001
TV	0.0191	0.002	12.70	< 0.0001
radio	0.0289	0.009	3.24	0.0014
TV×radio	0.0011	0.000	20.73	< 0.0001

β_0, β_1, β_2, β_3

- Increase TV by \$1,000

Increases sales by

$$(\beta_1 + \beta_3 X_{radio}) \cdot 1000$$

$$= (0.0191 + 0.0011\, X_{radio}) \cdot 1000$$

$$= 19 + 1.1 \cdot X_{radio}$$

Increasing radio by 1000

$$(\beta_2 + \beta_3 X_{TV}) \cdot 1{,}000 = 29 + 1.1 \cdot X_{TV}$$

Hierarchy principle

$X_1 \quad X_2 \quad X_1X_2 \quad X_3$

Sometimes p-value for interaction term is very small, but associated main effects are not.

The hierarchy principle:

If we include the interaction term, we should include the main effects no matter what

Interaction term for qualitative variables

Without interaction term

$$x_i = \begin{cases} 1 & \text{if student} \\ 0 & \text{if not} \end{cases}$$

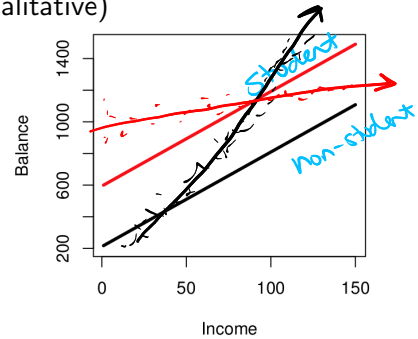
For credit data set:

Predict balance using income (quantitative) and student (qualitative)

$$\texttt{balance}_i \approx \beta_0 + \beta_1 \cdot \texttt{income}_i + \begin{cases} \beta_2 & \text{if student} \\ 0 & \text{if not} \end{cases}$$

$$\approx \beta_1 \cdot \texttt{income}_i + \begin{cases} \beta_0 + \beta_2 & \text{if student} \\ \beta_0 & \text{if not} \end{cases}$$

slope var intercept



Interaction term for qualitative variables

With interaction term

For credit data set:

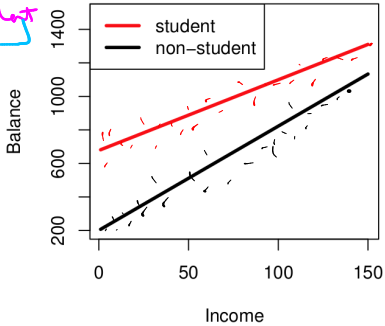
Predict balance using income (quantitative) and student (qualitative)

$\beta_0 + \beta_1 \text{ income} + \beta_2 \cdot \underbrace{\text{student}}_{0} + \beta_3 \underbrace{\text{income} \cdot \text{student}}_{0}$

$$\texttt{balance}_i \approx \underbrace{\beta_0}_{\text{int.}} + \underbrace{\beta_1}_{\text{slope}} \cdot \underbrace{\texttt{income}_i}_{\text{var}} + \begin{cases} \beta_2 + \beta_3 \cdot \texttt{income}_i & \text{if student} \\ 0 & \text{if not} \end{cases}$$

$$\approx \begin{cases} (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \cdot \texttt{income}_i & \text{if student} \\ \beta_0 + \beta_1 \cdot \texttt{income}_i & \text{if not} \end{cases}$$

int. slope var



Nonlinear relationships

Auto

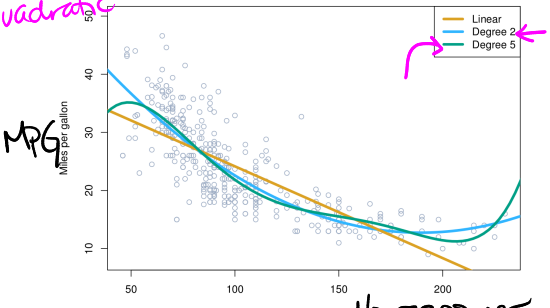
$\beta_0 + \beta_2 h^2$

$$\texttt{mpg} = \beta_0 + \beta_1 \cdot \texttt{horsepower} + \beta_2 \cdot \texttt{horsepower}^2 + \varepsilon$$

$\beta_3 h^3 + \beta_4 h^4 + \beta_5 h^5$

- Shape of data suggests quadratic
- Still linear b/c doing multi-linear reg w/
  $X_1 = \text{horsepower}$
  $X_2 = \text{horsepower}^2$
- Could add more terms
  run risk of overfitting

Polynomial Regression (More later)



	Coefficient	Std. error	t-statistic	p-value
Intercept	56.9001	1.8004	31.6	< 0.0001
horsepower	-0.4662	0.0311	-15.0	< 0.0001
horsepower^2	0.0012	0.0001	10.1	< 0.0001

Next time

- Fri:
  - Linear regression lab Zoom
  - HW #2 due. Covers:
    - 3.1 Linear regression (last friday)
    - 3.2 Multiple linear regression (today)
    - 3.3 Whatever we cover on Wednesday
- Monday:
  - 4.1-4.3 Intro to classification
  - In-person (unless something changes) Erickson 226
  - Zoom link available for medical need (i.e. covid exposure) or with permission of instructor. Send a slack message if you need to join via zoom.
  - No guarantee I can keep track of the chat on zoom